Modeling Employment Trends and AI-Driven White-Collar Job Displacement in Illinois

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This project proposes a data-driven analysis of employment trends in the State of Illinois from 1990 to the present, followed by a forward-looking simulation of potential white-collar job displacement driven by artificial intelligence (AI). We'll start by creating a DataFrame from the Illinois Department of Employment Security for employment in industries across Illinois (5th overall in agriculture, 6th in overall tech volume, 1st for AI Infrastructure) to visualize the economy of Illinois from 1990-present. Accessing the historical record, we'll establish the cause for job loss at recession points and use a Global AI Jobs 2026 Dataset https://www.kaggle.com/datasets/datascikhan/global-ai-jobs-2026 with regression analysis to determine a multiple for how many people in white collar jobs can be replaced by AI. Once we've identified these from a list of searchable job titles (Labor Force Statistics from the Current Population Survey) for the specific careers at risk we'll create a DataFrame with current job numbers from Illinois using the Bureau of Labor Statistics API (from ECON 8320) to visualize a recession scenario in December 2030 specific to white collar employment, provided there is no government intervention.